

On-Chain vs Off-Chain Data

Storing all data directly on the blockchain is not practical due to several limitations. First, it is **expensive**, as every piece of data stored incurs high gas costs. Second, blockchain data is **publicly accessible**, which raises privacy concerns. Finally, **large and sensitive data** cannot be efficiently stored on-chain due to storage constraints.

To address these issues, a **hybrid storage model** is used. In this approach, only critical and minimal data is stored on-chain, while larger and descriptive data is stored off-chain.

On-Chain Data

The blockchain (e.g., Ethereum smart contracts) stores essential information required for the application logic:

- Auction ID and seller address
- Highest bid and bidder details
- Auction deadline and status

Off-Chain Data

Heavy and descriptive data is stored externally:

- Item name and description
- Documents
- Images

Instead of storing full data on-chain, only a **Content Identifier (CID)** is stored. The actual data is securely stored off-chain, ensuring efficiency and scalability.

IPFS Integration (Using Pinata)

What is IPFS?

IPFS (InterPlanetary File System) is a **decentralized storage system** where:

- Data is identified using a **CID (hash)** instead of a location
- Files are distributed across a peer-to-peer network

Why Use IPFS?

IPFS is preferred for off-chain storage because:

- It is **gas-efficient**, avoiding large on-chain storage costs
- It ensures **immutability**, meaning data cannot be tampered with
- It is **decentralized**, removing reliance on a single server

Role of Pinata

Pinata is used as an IPFS pinning service to simplify file uploads and management.

Workflow in the Project

1. Upload item data (images, descriptions) to Pinata
2. Pinata returns a unique **CID**
3. Store this CID in the smart contract:

```
string metadataCID;
```

4. The frontend retrieves data using the gateway:

```
https://gateway.pinata.cloud/ipfs/<CID>
```

Security Considerations

- No personal or sensitive data is stored on-chain
- Only the **CID (reference)** is stored on the blockchain
- Actual data remains securely stored on IPFS

This architecture ensures a balance between **security, cost-efficiency, and decentralization**, making it ideal for blockchain-based applications like decentralized auctions.